Energetic Chemical Linkages

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Background

Chemical linkages combine rigid rods, movable joints, and binding sites. Binding changes which motions are possible; constructions include logic, catalysts, and a fuel-driven motor [1,2]. Thermodynamic binding networks suggest how to add preferences: $E(C) = -w\,b(C) - p(C)$ rewards bonds $b(C)$ and separate complexes $p(C)$; lower energy is more favorable [3]. Can a similarly simple score capture both binding and mechanical freedom?

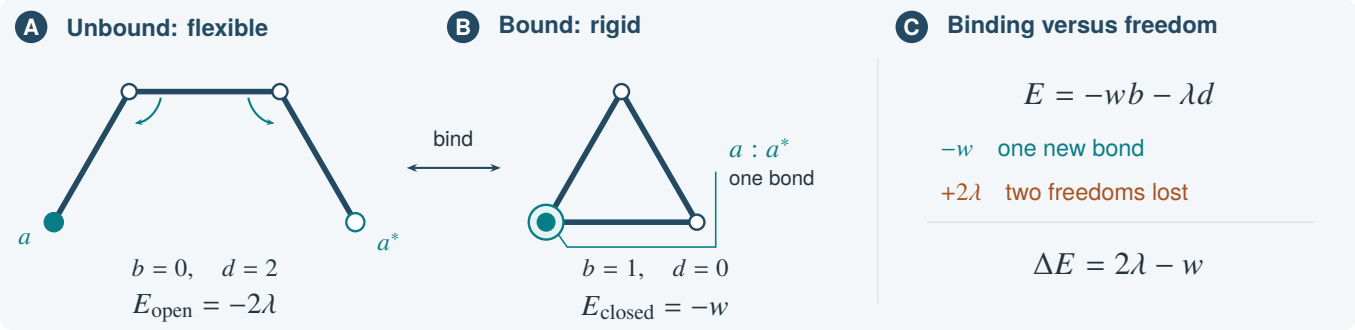


Figure 1. The same three equal-length rods before and after their end joints bind. Here d counts internal degrees of freedom, ignoring whole-body motion; it is an entropy proxy, not physical entropy.

The problem

1. **Define an energetic model of chemical linkages.** Specify configurations, allowed transformations, and an energy function. A starting point is $E(C) = -w\,b(C) - \lambda\,d(C)$, where $w, \lambda > 0$, bonds give an *enthalpic reward*, and independent mechanical degrees of freedom $d(C)$ provide an *entropy proxy*. Both rewards lower energy. Define how freedom is counted, including separate complexes. Explain the approximation or propose a better one; state how energy affects allowed or preferred transformations.
2. **Use the model to drive a motor.** Give a thermodynamically driven implementation of the motor in *Molecular Machines from Topological Linkages*, Section 5.2 [1]. Explain how fuel binding, splitting, and product release favor a mechanical cycle. Track motor, fuel, and products together: the motor may reset, but the complete system must record consumed fuel. Explain why intended motion is favored over reversal or getting stuck. A claim of probabilistic bias also requires a transition rule, not just state energies.

A starting point

First check the binding–flexibility tradeoff in Figure 1: closure is favorable in this toy score when $w > 2\lambda$. Then draw one motor cycle, label every state with bond counts, freedom, and energy, and search for suitable parameters. A useful result is one fully analyzed fuel-consumption cycle, or a counterexample showing why a proposed score cannot drive it. AI can help enumerate states, but the model and conclusion should be understandable on paper.

References

[1] K. Breik, A. Luchsinger, and D. Soloveichik. Molecular Machines from Topological Linkages. *DNA 27*, LIPIcs 205, 7:1–7:20, 2021. See Sections 5.1–5.3.

[2] T. Omabegho. Allosteric linkages that emulate a molecular motor enzyme. *bioRxiv*, 2021. doi:10.1101/2021.04.20.440673 (version 1).

[3] K. Breik, C. Chalk, D. Doty, D. Haley, and D. Soloveichik. Programming Substrate-Independent Kinetic Barriers with Thermodynamic Binding Networks. *IEEE/ACM Trans. Comput. Biol. Bioinform.* 18(1):283–295, 2021.